

Name & ID: Syed Ali Nisar Shah (ss07136) & Syed Muhammad Asad Ali (sa09475)

Date: October 10, 2025

Lab 7: QPSK modulation

Objective

The objective of this lab is to observe the Quadrature Phase Shift Keying (QPSK) and Binary Phase Shift Keying modulation and demodulation using Simulink.

Simulink Model of QPSK Modulator and Demodulator

Task 1: Standard QPSK Simulation

The Simulink model of QPSK modulator and demodulator is shown below

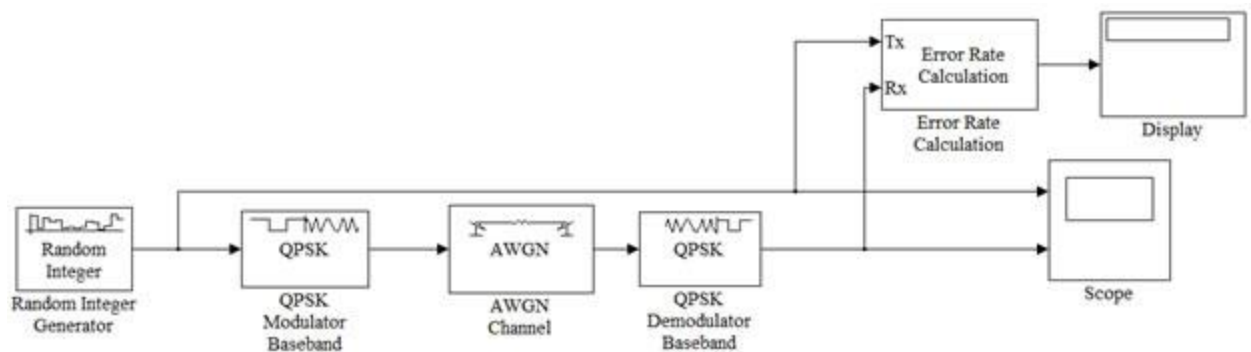


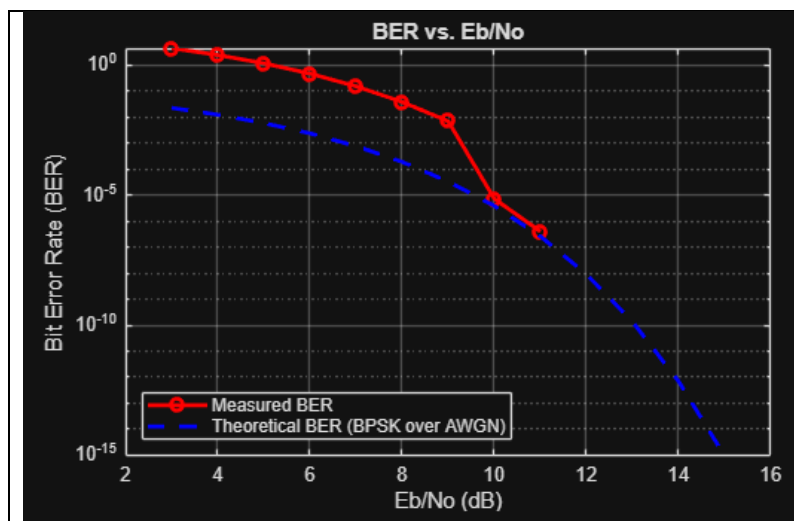
Figure 1. QPSK Simulation Model

- Build the Simulink model shown in Figure 1.
- Double-click on the Random Integer Generator and adjust the set size to a proper value (Remember that the input to the QPSK modulator should be either 0, 1, 2, or 3). What should be the value of M? $M = 4$
- In the Random Integer Generator block, set the *Sample Time* to $1e-6$ (i.e. $1 \mu s$) and the *Samples per frame* parameter to 1024.
- In the AWGN block, set the *Symbol period* parameter to $1e-6$ (i.e. $1 \mu s$) and the *Number of bits per symbol* parameter to 2 (since QPSK uses 2 bits per symbol).
- For the Error Rate Calculation block, set the *Output data* field to "port" so you can connect the Display block.
- The Display Block will show you three values. The first value is the BER, the second value is the number of incorrect bits, and the third value is the total number of bits received.
- Set the simulation time to 10 seconds.
- In both QPSK Modulator and Demodulator blocks, set the *Constellation ordering* to Gray. Take a note of the constellation points and attach both plots.

- In this part, you will adjust the value of E_b/N_0 in the AWGN block, starting from 3, incrementing by 1 every step, and ending at 15, and observe the error rate displayed in the Display block. Make a table recording the value of E_b/N_0 and the corresponding BER and attach below.

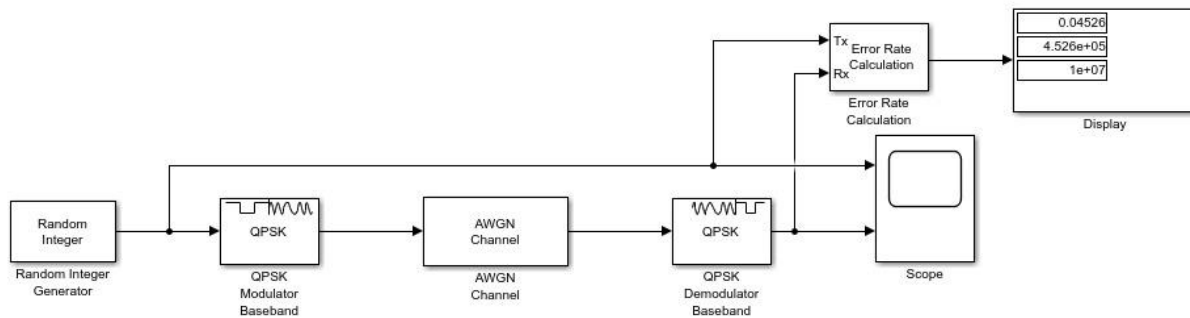
E_b/N_0	BER
3	4.531
4	2.48
5	1.189
6	0.4781
7	0.1556
8	0.03865
9	0.00707
10	0.0000073
11	0.0000004
12	0
13	0
14	0
15	0

- Plot BER vs. E_b/N_0** and compare with the theoretical values. **Comment** on the results below.



The simulated QPSK system performs very close to theory, with small deviations at low SNR due to simulation or practical factors.

Task 1 model:



Task 2: Comparing QPSK and BPSK

In this experiment, you will learn about the trade-off of using QPSK over BPSK.

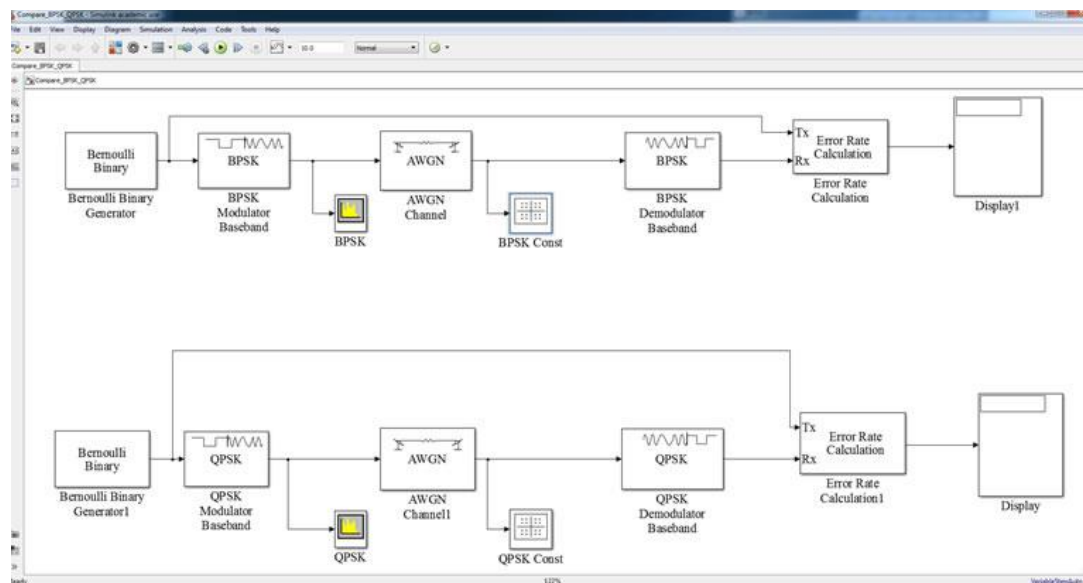


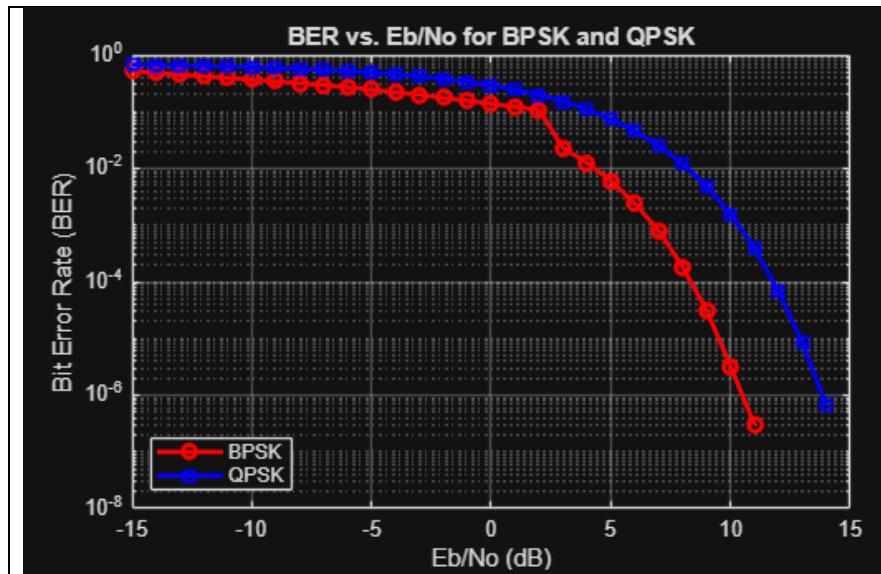
Figure 2. QPSK vs. BPSK

- Build the Simulink model shown in Figure 2.
- Double click on the Bernoulli generator for the BPSK part. Set the sample time to 1e-6 and the Samples per frame to 1024.
- Double click on the Bernoulli generator for the QPSK part. Set the sample time to 0.5e-6 and the Samples per frame to 1024.
- For the QPSK Modulator and Demodulator blocks, use Gray Constellation ordering.
- Set the simulation time to 10 seconds.
- Adjust the value of E_b/N_0 in the AWGN block, starting from 3, incrementing by 1 every step, and ending at 15, and observe the error rate displayed in the Display block. Make a table recording the value of E_b/N_0 and the corresponding BER and attach below. Run

the simulation. observe and record the bit error rate and the number of transmitted samples from the Display block for both schemes. **Explain** your observations.

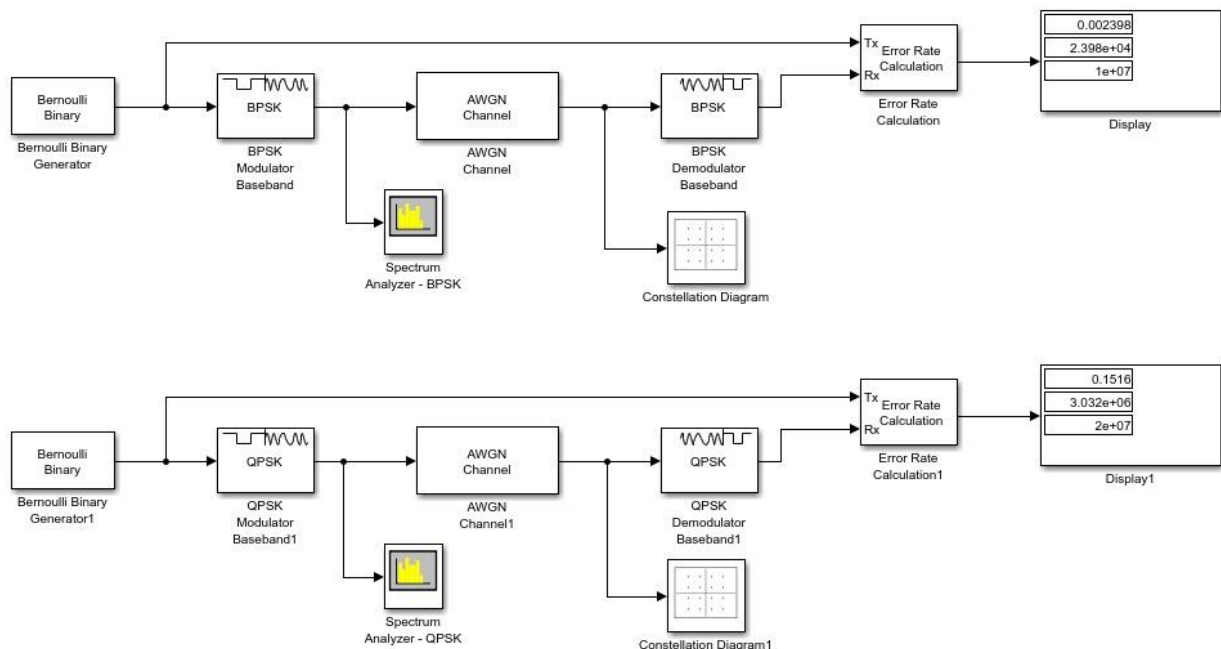
Eb/No	BPSK BER	QPSK BER
-15	0.51765	0.6744
-14	0.4793	0.6646
-13	0.4469	0.6537
-12	0.4188	0.6411
-11	0.3925	0.6269
-10	0.3665	0.6105
-9	0.3417	0.5922
-8	0.3173	0.5713
-7	0.2931	0.5477
-6	0.2688	0.5213
-5	0.2443	0.4917
-4	0.2203	0.4584
-3	0.1972	0.4217
-2	0.1756	0.3813
-1	0.1559	0.3379
0	0.1380	0.292
1	0.1220	0.2446
2	0.1079	0.1974
3	0.02284	0.1517
4	0.01251	0.1098
5	0.00601	0.07398
6	0.0024	0.04549
7	0.000791	0.02502
8	0.0001803	0.01197
9	0.0000311	0.0048
10	0.0000032	0.001556
11	0.0000003	0.0003919
12	0	0.0000672
13	0	0.00000835
14	0	0.00000065
15	0	0

- **Plot** BER vs. Eb/No and compare with the theoretical values. **Comment** on the results below.



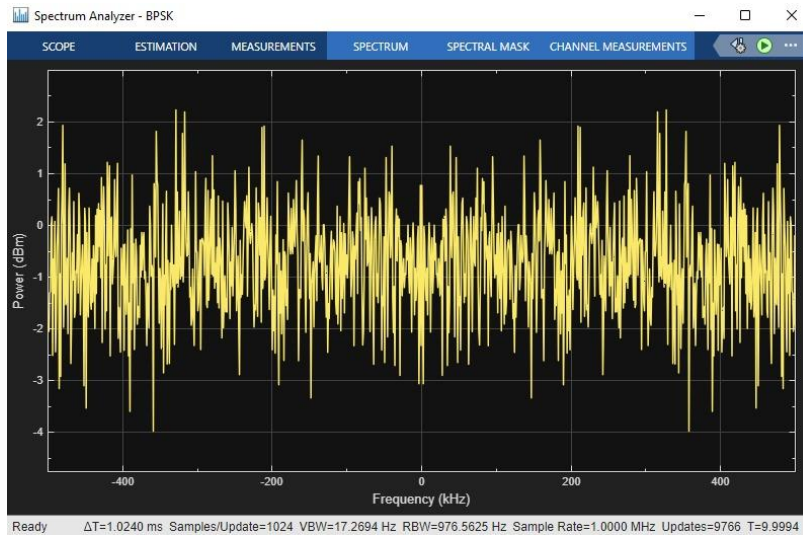
BPSK and QPSK have equivalent BER performance over AWGN channels, especially at the lower signal-to-noise ratio. QPSK achieves the same BER while doubling the data rate (two bits per symbol), making it more bandwidth-efficient.

Task 2 model:

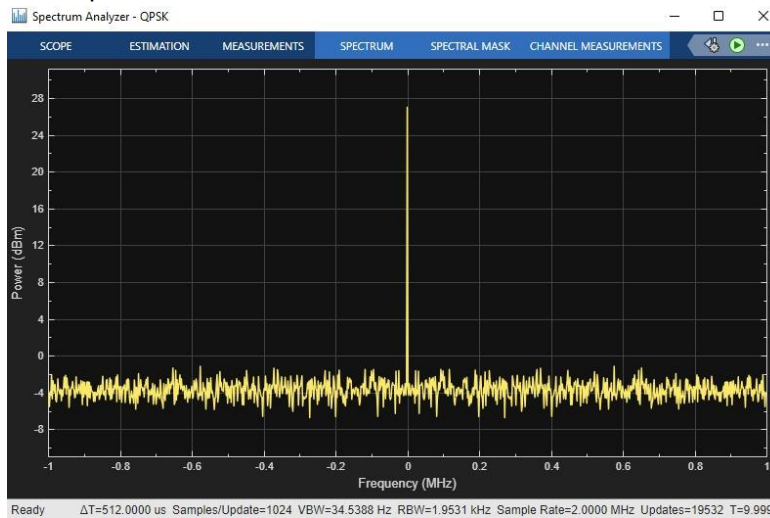


Task 2 plots:

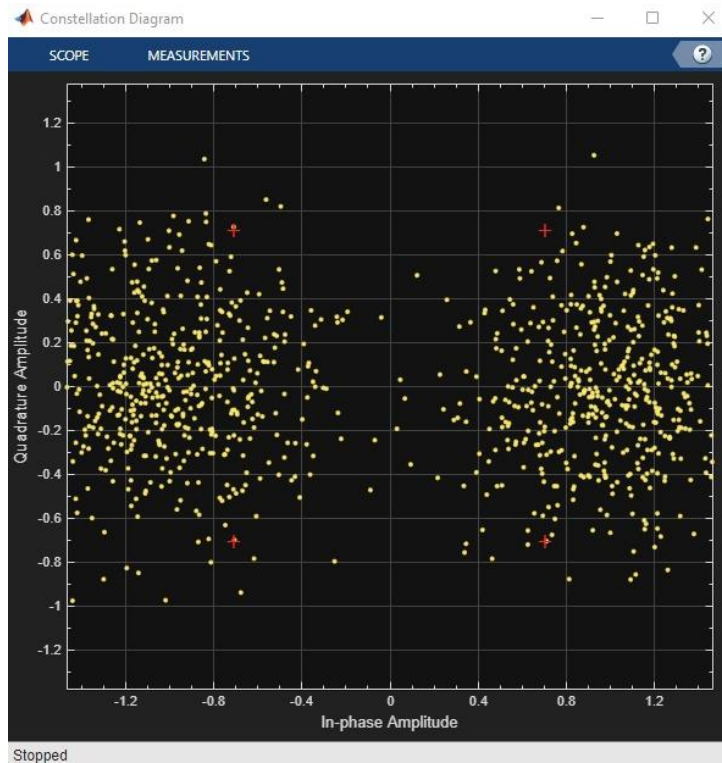
1. BPSK spectrum:



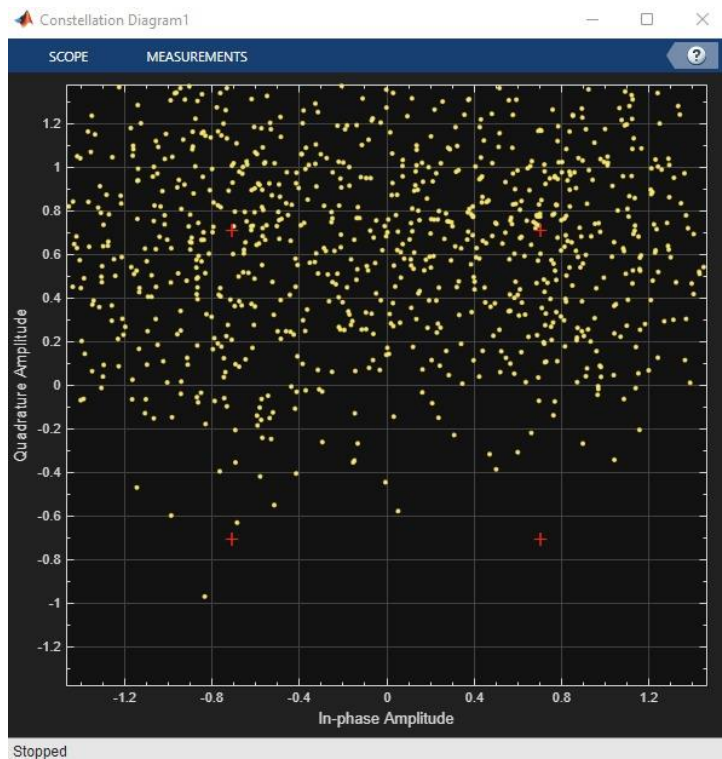
2. QPSK spectrum:



3. BPSK constellation:



4. QPSK constellation:



*Attach Simulink model of all tasks in the submitted word manual.

Task 3: A Music File Transmission with QPSK

In this task, you will adjust the value of SNR in the AWGN block and observe the quality of the music. Choose at least three values of SNR (high, mid, low) and comment on the quality of the music.

If the music file length is 62 second, so set the Simulation Time to 62 seconds.

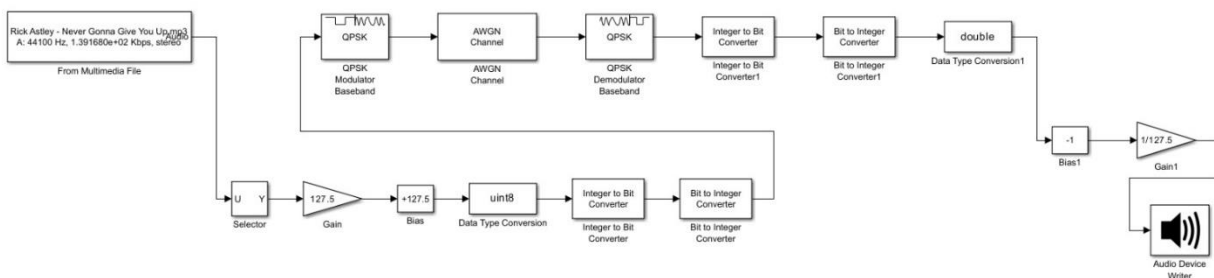
Note: You have to upload Simulink file and music file with the Lab Report on LMS.

At low SNR of 3: The playback has a lot of white noise that dominates the audio output. The music is barely understandable.

At mid SNR of 10: The playback still has faint white noise and random pops, but one can understand what music is being played.

At high SNR of 20: Most of the white noise is gone and we get almost the same quality in playback as the original audio we gave as input.

Simulink model:



Name & ID: _____

Date: _____

EE-322L Analog and Digital Communications Lab

Fall 2025

Habib University

Dhanani School of Science & Engineering



LAB 7 QPSK Modulation

Lab #7 Marks distribution:

		LR1=20	LR3=10	LR4=30	LR5=30	AR4=20
In-Lab Tasks	Task 1	/10	/05	/10	/10	/20
	Task 2	/05	/05	/10	/10	
	Task 3	/05		/10	/10	
Marks Obt.	/100					

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR1	Neat and Clean Circuit layout	Components are wired and didn't show neat and clean connections and minimal efforts shown.	Components are wired with untidy connection and didn't show neat and clean connections	Few but not all Components are wired with neat and clean connections	Complete components are wired with neat and clean tight connections. Given task completed in due time.

LR3	Troubleshooting	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included	Measurements are somewhat inaccurate and very imprecise. Observations are incomplete or recorded in a confusing way. Major errors using symbols, units and significant digits	Measurements are mostly accurate. Observations are generally complete. Minor errors using symbols, units and significant digits	Measurements are both accurate and precise. Observations are very thorough. Includes appropriate symbols, units and significant digits and completed in due time.
LR5	Results & Plots	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions, units missing or incorrect, etc.	Most figures, graphs, tables OK, some still missing some important or required features	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved	All figures, graphs, tables are correctly drawn and contain titles/captions.
LR8	Contribution	No participation towards the group tasks	Slight participation towards the group tasks	Substantial participation towards the group tasks	Outstanding participation towards the group tasks
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after 1 week of due date